

Objective

To configure and analyze DHCP services in Cisco Packet Tracer, specifically focusing on the DORA process (Discover, Offer, Request, Acknowledge) and validating IP address assignment in both GUI and CLI environments.

Tool Used

Cisco Packet Tracer.

Methodology

The lab is conducted in two main phases:

1. **GUI-based Configuration:** A topology was created with a Server and two PCs connected to a switch. The DHCP service was enabled on the server to dynamically assign IP addresses to the PCs. Packet simulation was used to capture and analyze the DHCP handshake process.
2. **CLI-based Configuration:** A router-based topology was implemented. The router interface (GigabitEthernet0/0) was configured with a static IP address, and a DHCP pool was created via CLI, including address exclusion and necessary network parameters (default router, DNS).

Observations

Analysis of the DHCP process (DORA):

- **Discover:** The DHCP client broadcasts a Discover packet on the network to locate available DHCP servers.
- **Offer:** Upon receiving the Discover, the server responds with a DHCP Offer packet, proposing an IP configuration (IP address, subnet mask).
- **Request:** The client accepts the offer by sending a DHCP Request packet, formally asking to lease the offered IP address.
- **Acknowledge:** Finally, the DHCP server sends a DHCP Acknowledge (ACK) packet, confirming the lease and finalizing the configuration.

Captured data in Figure 5 (PDU details) and Figure 6 (Simulation Panel) confirm this DORA sequence. Figure 6 also shows background Spanning Tree Protocol (STP) traffic, which is typical for network switches, but confirms the DHCP handshake is captured successfully.

Images

Using GUI:

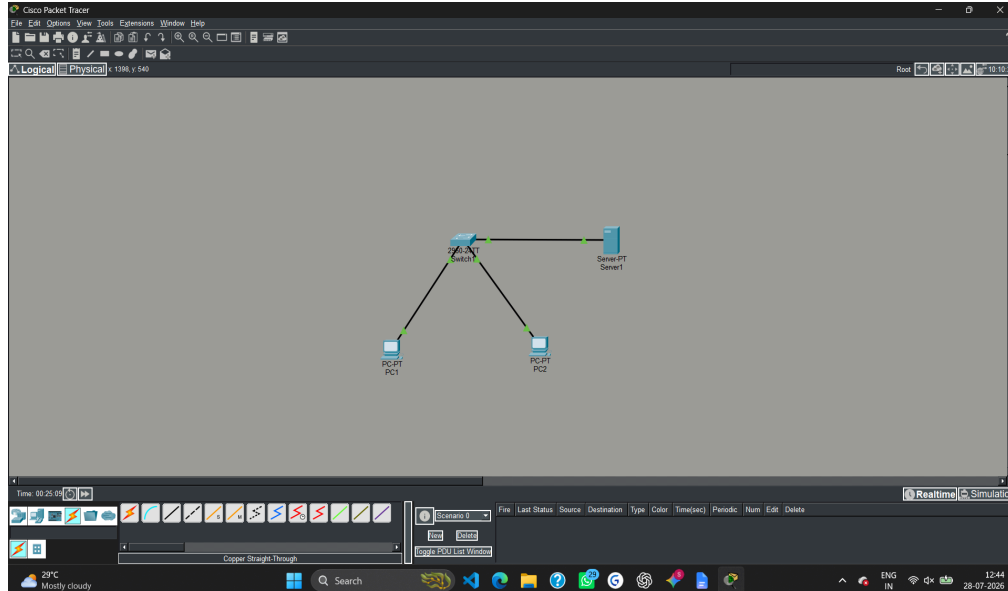


Figure 1: Network topology featuring a switch, server, and two PCs in Cisco Packet Tracer.

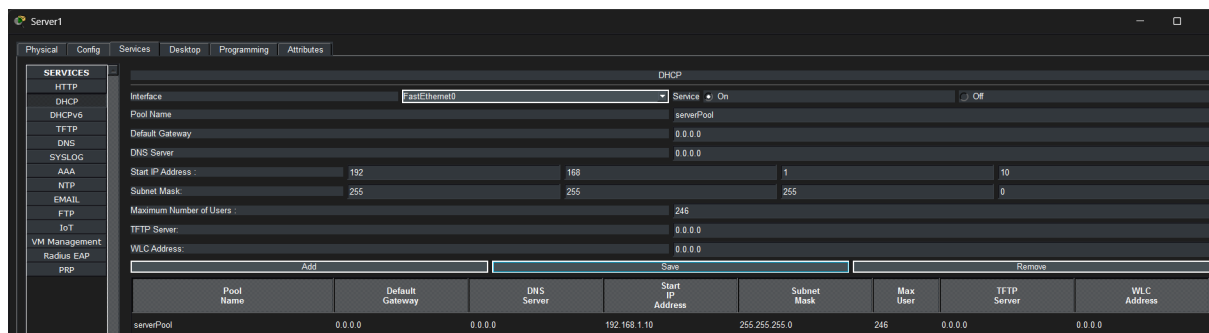


Figure 2: DHCP service configuration settings on Server1.

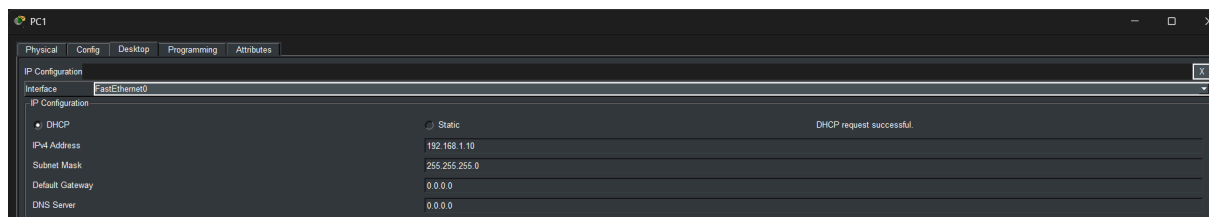
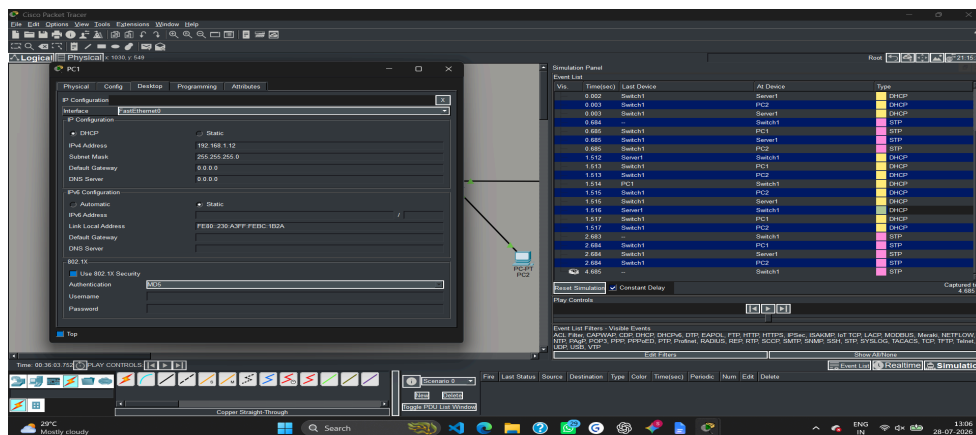
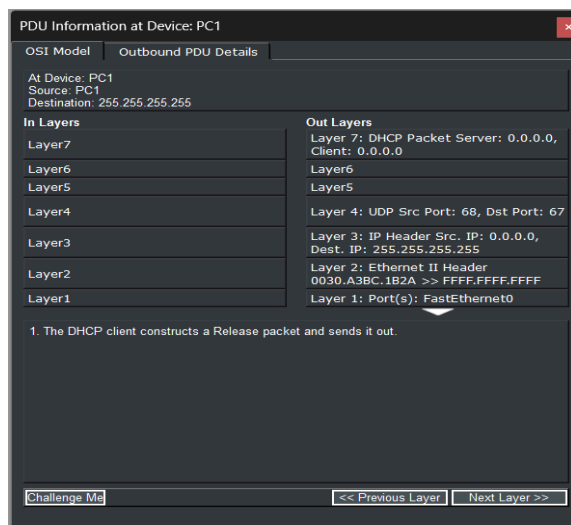
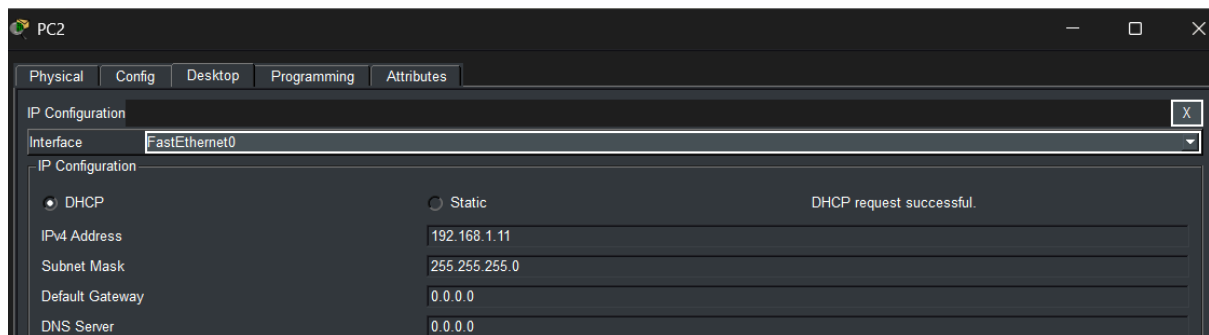


Figure 3: Successful DHCP IP configuration on PC1.



Using Router's CLI:

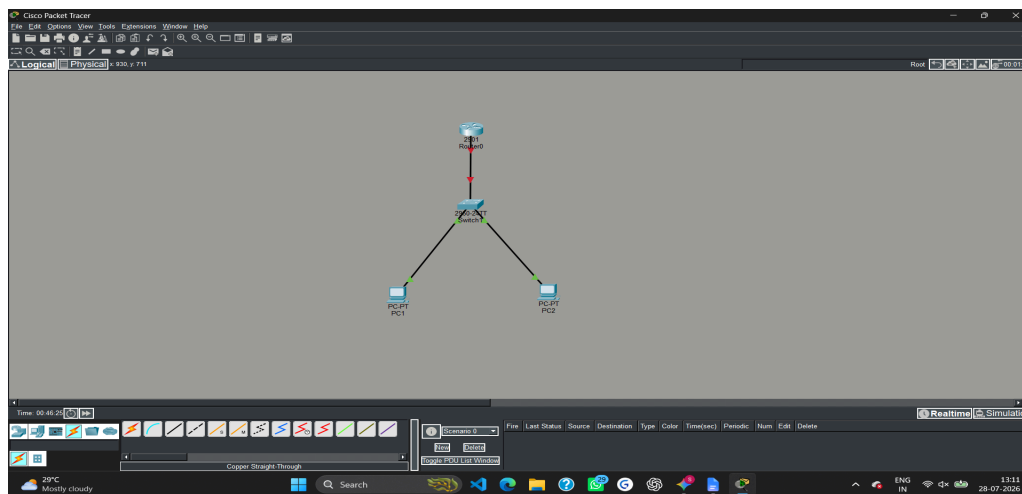


Figure 7: Updated network topology including a router connected to the switch and PCs.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#
```

Figure 8: CLI commands for configuring the router's GigabitEthernet0/0 interface.

```
Router(config)#ip dhcp excluded-address 192.168.1.1 192.168.1.9
Router(config)#ip dhcp pool LAN_POOL
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#domain-name example.com
Router(dhcp-config)#exit
Router(config)#
```

Figure 9: CLI commands for setting up the DHCP pool named 'LAN_POOL'.

```
Router#show ip dhcp binding
IP address      Client-ID/      Lease expiration    Type
                Hardware address
192.168.1.10    0030.A3BC.1B2A  --                  Automatic
192.168.1.11    0030.F2B7.C597  --                  Automatic
Router#show ip dhcp pool LAN_POOL
Pool LAN_POOL :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 2
  Excluded addresses              : 1
  Pending event                  : none

  1 subnet is currently in the pool
  Current index    IP address range    Leased/Excluded/Total
  192.168.1.1     192.168.1.1 - 192.168.1.254    2 / 1 / 254
Router#
```

Figure 10: Output showing the active DHCP bindings and pool status on the router.

Conclusion

DHCP services were successfully implemented using both GUI-based server settings and CLI-based router configuration. The simulation successfully captured the DORA process, verifying that IP addresses were assigned correctly to clients in both test scenarios.

Skills Learned

- Configuring DHCP server pools and address exclusions.
- Router CLI commands for network interface and DHCP setup.
- Practical network simulation using Cisco Packet Tracer.
- Packet analysis, including inspecting PDU layers and interpreting DHCP handshake sequences.